

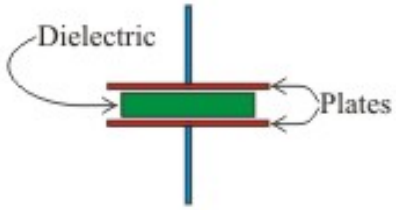


Instrumentación Electrónica

Área de Electrónica
Departamento de Física Aplicada

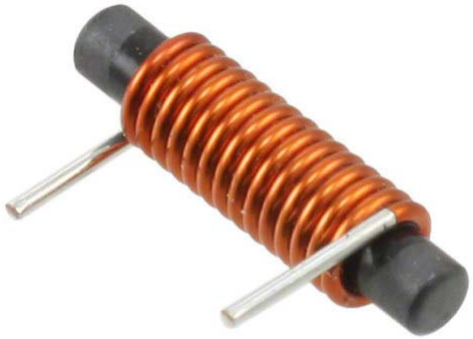
Transformada de Laplace

$$F(s) = \int_0^{\infty} f(t)e^{-st}dt \qquad f(t) = \frac{1}{2\pi j}\lim_{T\to\infty}\int_{\gamma-jT}^{\gamma+jT} F(s)e^{st}\,dt$$



$$i(t) = \frac{dq(t)}{dt} = C\frac{dv(t)}{dt}$$

$$Z_C(\omega) = \frac{V(\omega)}{I(\omega)} = \frac{1}{j\omega C} = \frac{1}{Cs}$$



$$v(t) = L\frac{di(t)}{dt}$$

$$Z_L(\omega) = \frac{V(\omega)}{I(\omega)} = j\omega L = Ls$$

Transformada de Laplace de una señal

$$\mathcal{L}[f(t)] = F(s) = \int_0^\infty e^{-st}\,dt[f(t)] = \int_0^\infty f(t)e^{-st}\,dt$$

Función en el dominio del tiempo

LT en el dominio del tiempo

$$\mathcal{L}^{-1}[F(s)] = f(t) = \frac{1}{2\pi j}\int_{c-j\infty}^{c+j\infty} F(s)e^{st}\,ds, \qquad \text{for } t > 0$$

LT inversa en el dominio del tiempo

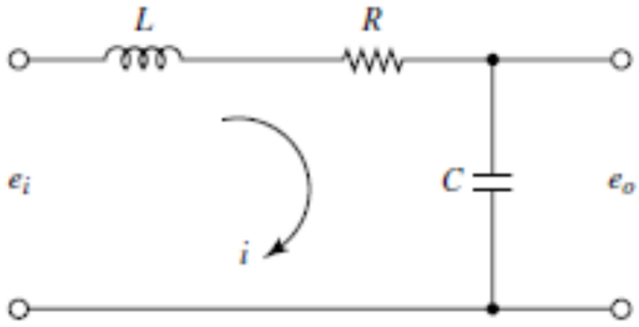
Transformada de Laplace: Propiedades

1	$\mathcal{L}[Af(t)] = AF(s)$
2	$\mathcal{L}[f_1(t) \pm f_2(t)] = F_1(s) \pm F_2(s)$
3	$\mathcal{L}_{\pm}\left[\frac{d}{dt}f(t)\right] = sF(s) - f(0\pm)$
4	$\mathcal{L}_{\pm}\left[\frac{d^2}{dt^2}f(t)\right] = s^2F(s) - sf(0\pm) - \dot{f}(0\pm)$
5	$\mathcal{L}_{\pm}\left[\frac{d^n}{dt^n}f(t)\right] = s^nF(s) - \sum_{k=1}^{n} s^{n-k}f^{(k-1)}(0\pm)$ where $f^{(k-1)}(t) = \frac{d^{k-1}}{dt^{k-1}}f(t)$
6	$\mathcal{L}_{\pm}\left[\int f(t)\,dt\right] = \frac{F(s)}{s} + \frac{1}{s}\left[\int f(t)\,dt\right]_{t=0\pm}$
7	$\mathcal{L}_{\pm}\left[\int\cdots\int f(t)(dt)^n\right] = \frac{F(s)}{s^n} + \sum_{k=1}^{n}\frac{1}{s^{n-k+1}}\left[\int\cdots\int f(t)(dt)^k\right]_{t=0\pm}$
8	$\mathcal{L}\left[\int_0^t f(t)\,dt\right] = \frac{F(s)}{s}$
9	$\int_0^{\infty} f(t)\,dt = \lim_{s\to 0} F(s)$ if $\int_0^{\infty} f(t)\,dt$ exists

Transformada de Laplace: Aplicación al análisis de circuitos

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = e_i$$

$$\frac{1}{C}\int i\,dt = e_o$$



Assuming zero initial conconditions

$$LsI(s) + RI(s) + \frac{1}{C}\frac{1}{s}I(s) = E_i(s)$$

$$\frac{1}{C}\frac{1}{s}I(s) = E_o(s)$$

$$\longrightarrow \quad \frac{E_o(s)}{E_i(s)} = \frac{1}{LCs^2 + RCs + 1}$$

Serie de Fourier de una onda periódica

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(2\pi nft) + \sum_{n=1}^{\infty} b_n \sin(2\pi nft)$$

$$a_0 = \frac{1}{T}\int_{-T/2}^{+T/2} x(t)dt$$

$$a_n = \frac{2}{T}\int_{-T/2}^{+T/2} x(t)\cos\left(\frac{2\pi nt}{T}\right)dt$$

$$b_n = \frac{2}{T}\int_{-T/2}^{+T/2} x(t)\sin\left(\frac{2\pi nt}{T}\right)dt$$

Transformada de Fourier de una función

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} f(t) e^{-j2\pi f t} dt$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{+j\omega t} d\omega$$

Transformada de Fourier: Tabla funciones de interés

No.	$x(t)$	$X(\omega)$	
1	$e^{-at}u(t)$	$\frac{1}{a+j\omega}$	$a > 0$
2	$e^{at}u(-t)$	$\frac{1}{a-j\omega}$	$a > 0$
3	$e^{-a\|t\|}$	$\frac{2a}{a^2+\omega^2}$	$a > 0$
4	$te^{-at}u(t)$	$\frac{1}{(a+j\omega)^2}$	$a > 0$
5	$t^n e^{-at}u(t)$	$\frac{n!}{(a+j\omega)^{n+1}}$	$a > 0$
6	$\delta(t)$	1	
7	1	$2\pi\delta(\omega)$	

Transformada de Fourier: Tabla funciones de interés

No.	$x(t)$	$X(\omega)$
8	$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$
9	$\cos\omega_0 t$	$\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$
10	$\sin\omega_0 t$	$j\pi[\delta(\omega+\omega_0)-\delta(\omega-\omega_0)]$
11	$u(t)$	$\pi\delta(\omega)+\dfrac{1}{j\omega}$
13	$\cos\omega_0 t\,u(t)$	$\dfrac{\pi}{2}[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]+\dfrac{j\omega}{\omega_0^2-\omega^2}$
14	$\sin\omega_0 t\,u(t)$	$\dfrac{\pi}{2j}[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]+\dfrac{\omega_0}{\omega_0^2-\omega^2}$
17	$\text{rect}\left(\dfrac{t}{\tau}\right)$	$\tau\,\text{sinc}\left(\dfrac{\omega\tau}{2}\right)$